HEPS target UI design

Hospital Emergency Priority System · light theme · for implementation in Antigravity

How to use this document

Each following page shows one screen of the target design with build notes. Treat the pictures as the target UI. Match layout, colours, spacing, component shapes and hierarchy as closely as JavaFX allows. Where a screen is not shown, follow the same design language. Keep the existing min-heap, rule engine, AI orchestrator and JDBC layers; only the workflow, data model and UI change.

Design tokens

Canvas #F6F8FB	Surface #FFFFFF	Border #E5EAF1	Text #0F172A	Brand teal #0D9488	Teal tint #F0FDFA	Critical #B91C1C on #FEE2E2

High #C2410C on #FFEDD5	Medium #A16207 on #FEF9C3	Low #15803D on #DCFCE7

Rules

Light theme everywhere, including the sidebar, dialogs and banners. Font: Inter (13px body, 24px page title, 30px KPI numbers). Radius 12 for cards, 10 for inputs and buttons, 999 for pills. 1px borders, very soft shadows only. Primary button is solid teal; secondary is white with a border; no black buttons. Icons: Ikonli Material Design or Tabler style line icons, never emoji. Priority is always shown by colour and text together. Minimum window 1280 by 720 with no clipped columns.

Screen index (workflow order)

1. Command center
2. Login
3. Register patient · step 1: Identity
4. Register patient · step 2: Triage assessment
5. Register patient · step 3: Decision and routing
6. Triage ticket dialog
7. Emergency queue
8. Bay board
9. Doctor workspace
10. Orders and results
11. Disposition and discharge
12. Patient records
13. Alerts center
14. Reports and analytics
15. Admin
16. Mass casualty mode

01

Command center

Operations · live overview of the whole ED

HEPS
Emergency dept

Operations
- Command center
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- Emergency queue
- Bay board
- Mass casualty

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PS **Priya S.**
Triage nurse

Command center

Search token or UHID · 14:32 · 3

Waiting
6
avg 12 min to triage

Critical
2
in resus bays

SLA breaches
1
2 near limit

Bed use
9/12
75% occupied

1 Registered › 2 Triaged › 3 Waiting › 5 In treatment › 2 Disposition

Next patient to treat · Critical · Bay 1 · Resus · Target: immediate

TK-2026-0142 **Ramesh Kumar** 58 · Male · UH-2026-000891

SpO2 88% · HR 134 · BP 150/95 · NEWS2 9 · Allergy: penicillin

Chest pain radiating to left arm, sweating

Print slip · Begin treatment

Waiting queue

Ranked by min-heap priority, live

View full queue

#	Patient	Priority	Bay · doctor	Wait vs SLA
1	**Ramesh Kumar** TK-0142 · 58 M · chest pain	Critical	**Bay 1 · Resus** Dr. Mehta	3m / immediate
2	**Anita Joshi** TK-0143 · 34 F · leg fracture	High	**Bay 3 · Acute** Dr. Rao	12m / 15m
3	**Kavita Sundaram** TK-0140 · 45 F · fever	Medium	**Bay 4 · Obs** Unassigned	28m / 60m
4	**Tanya Kapoor** TK-0145 · 12 F · ankle sprain	Low	**Bay 6 · Fast track** Unassigned	18m / 120m

Build notes

- KPI strip: waiting, critical, SLA breaches, bed occupancy. Refresh every 15 seconds on a background thread.
- Pipeline strip shows patient counts per workflow stage: Registered, Triaged, Waiting, In treatment, Disposition.
- Next patient card uses PriorityQueue.peek(); left border colour follows priority; vitals chips flag abnormal values; allergy shown in red.
- Queue table is the top 4 to 6 rows of the heap. Each row shows live wait against the SLA target with a coloured bar.
- White sidebar, active item uses teal tint with a 3px accent bar. Nothing dark anywhere.

02 **Login**

Access · role-based sign in

HEPS

City General Hospital

Emergency care, prioritized in real time

Triage, bay allocation and treatment tracking for every patient, from arrival to discharge.



- Role-based access and audit trail
- Live SLA tracking and escalation

Sign in

Use your hospital staff account

Username

nurse.priya

Password

••••••••

Role

Triage nurse

Sign in

Demo accounts

Receptionist | Triage nurse | Doctor | Charge nurse | Admin

Build notes

- Split layout: soft teal brand panel on the left (solid #F0FDFA, no dark colours), white sign-in card on the right.
- Roles: Receptionist, Triage nurse, Doctor, Charge nurse, Admin. Passwords stored with BCrypt. Auto logout after inactivity.
- Demo account chips fill the credentials for quick demos. Sidebar after login shows only screens allowed for the role.
- Every login and logout is written to the audit log.

03

Register patient · step 1: Identity

Arrival · demographics, arrival mode, MLC flag

Register patient

Search token or UHID · 14:32 · 3

1 Identity — 2 Triage assessment — 3 Decision

Returning patient found
Ramesh Kumar · UH-2026-000891 · last visit 02 Aug 2026 (fever)

Use this record

Patient details

Full name * Ramesh Kumar

Contact number * 98765 43210

Date of birth * 14 / 03 / 1968

Age 58 years

Gender * Male

Address 12, Lake View Road, Chennai

Blood group B+

Emergency contact

Name Sunita Kumar

Relation Spouse

Phone 98765 11122

Arrival

Walk-in | Ambulance | Referred | Police | Brought by relatives

☐ Medico-legal case (MLC)

Cancel

Continue to triage

Build notes

- 3-step wizard with numbered circles and a progress line; completed steps show a teal check.
- UHID is generated automatically (UH-2026-000123 format). Returning patients are detected by phone or name plus date of birth and show a blue banner with last visit.
- Fields: name, phone, DOB (age auto-calculated), gender, address, blood group, emergency contact, arrival mode chips, MLC checkbox.
- Inline validation in red under each field. Required fields marked with a red asterisk.

04

Register patient · step 2: Triage assessment

Triage · mandatory vitals, complaint, live decision support

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Emergency dept

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PS Priya S. Triage nurse

Register patient

Search token or UHID · 14:32 · 3

Identity — 2 Triage assessment — 3 Decision

Vital signs *

Ramesh Kumar · 58 M · UH-2026-000891

SpO2 (%)	Heart rate (bpm)	Temperature (F)
88 Critical	134 High	98.6
Systolic BP (mmHg)	**Diastolic BP (mmHg)**	**Respiratory rate (/min)**
150	95	26

GCS (3 to 15): 15

Pain score: 8 / 10

Chief complaint *

Chest pain | Breathlessness | Trauma | Stroke signs | Fever | Bleeding

Chest pain radiating to left arm, sweating, onset 30 minutes ago

Allergies: Penicillin

Known conditions: Diabetes, hypertension

Clinical decision support

NEWS2: 9

Rule engine
Critical
SpO2 under 90%

AI agents
High
Possible acute coronary syndrome

⚠ Disagreement detected. Defaulting to the higher acuity: Critical.

Back

Continue to decision

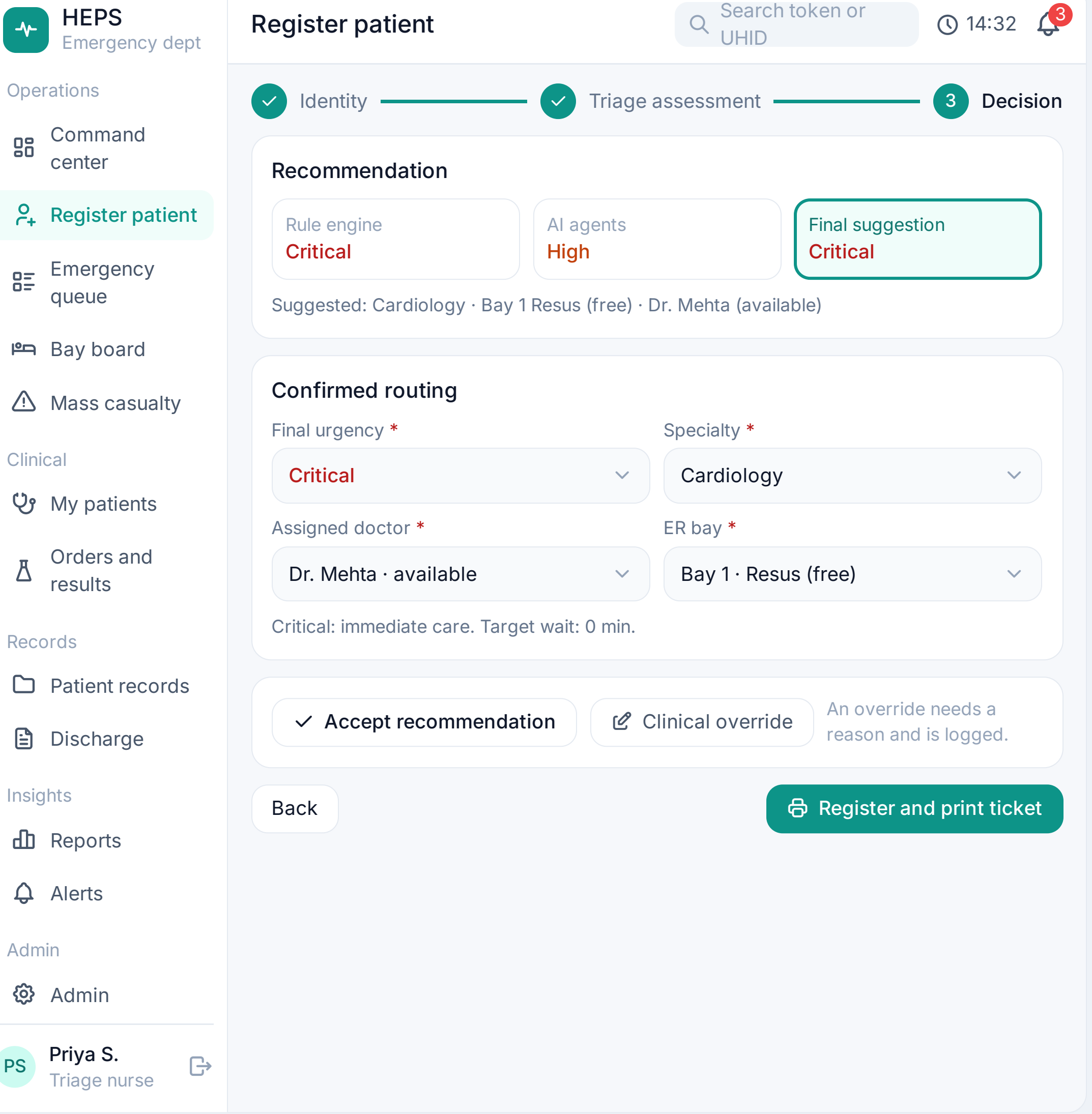
Build notes

- Vitals are mandatory. A nurse can skip only with Unable to record plus a reason, which is audited.
- Abnormal fields turn red or orange as the nurse types, using JavaTriageRuleEngine thresholds (SpO2 under 90 critical, HR over 130 critical, and so on).
- NEWS2 early warning score is calculated live from the vitals. GCS and pain slider are stored in the visit.
- Decision support compares Rule Engine and AI agents. If they disagree, show the amber banner and default to the higher acuity.

05

Register patient · step 3: Decision and routing

Triage · confirm acuity, department, doctor, bay



Build notes

- Only free bays of the right type and only available doctors with the matching specialty appear in the dropdowns. Occupied bays cannot be chosen.
- Accept recommendation or Clinical override. Override requires a reason and is written to the audit log with old and new value.
- Register and print ticket creates the visit, enqueues into the min-heap, reserves the bay, and opens the triage ticket dialog.
- Status moves REGISTERED to TRIAGED to WAITING through the state machine.

06

Triage ticket dialog

Triage · printable slip with barcode and QR

Triage ticket · TK-2026-0142

City General Hospital

Department of Emergency Medicine · Triage slip

Patient token

TK-2026-0142

UHID UH-2026-000891

Arrival

22 Oct 2026, 14:21

Critical · Target: immediate · Bay 1 · Resus

Allergy: penicillin

Patient	Ramesh Kumar
Age · sex	58 · Male · 98765****10
Department	Cardiology
Doctor	Dr. Mehta

Vitals baseline

SpO2 88% · HR 134 · BP 150/95 · Temp 98.6 F · RR 26 · NEWS2 9

Complaint

Chest pain radiating to left arm, sweating

* TK-2026-0142 *

Re-triage due by 14:36

Print ticket | Wristband | Close

Build notes

- Opens automatically after registration and from Queue or Records via Print triage slip. White dialog, 16px radius, light grey overlay.
- Content: hospital header, token, UHID, arrival time, priority pill with target SLA, bay, allergy alert in red, masked phone, vitals baseline with NEWS2, complaint.
- Code-128 barcode plus QR code (ZXing). Re-triage due time is shown. Wristband button prints a patient label.
- Printing uses JavaFX PrinterJob with a fallback preview.

07

Emergency queue

Waiting · heap-ordered patients with live SLA

HEPS
Emergency dept

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PS Priya S.
Triage nurse

Emergency queue

Search token or UHID · 14:32 · 3

Scan wait escalations

Search token, UHID, name · Priority: All · Specialty: All

#	Patient	Priority	Bay · doctor	Wait vs SLA
1	Ramesh Kumar TK-0142 · 58 M · chest pain	Critical	Bay 1 · Resus Dr. Mehta	3m / immediate
2	Anita Joshi TK-0143 · 34 F · leg fracture	High	Bay 3 · Acute Dr. Rao	12m / 15m
3	Harish Patel (escalated) TK-0139 · 71 M · abdominal pain	High	Bay 3 · Acute Unassigned	52m / 60m
4	Kavita Sundaram TK-0140 · 45 F · fever	Medium	Bay 4 · Obs Unassigned	28m / 60m
5	Farhan Akhtar TK-0144 · 26 M · back pain	Medium	Bay 4 · Obs Unassigned	9m / 60m
6	Tanya Kapoor TK-0145 · 12 F · ankle sprain	Low	Bay 6 · Fast track Unassigned	18m / 120m

Harish Patel selected · Re-triage · Print slip · Assign doctor · Begin treatme

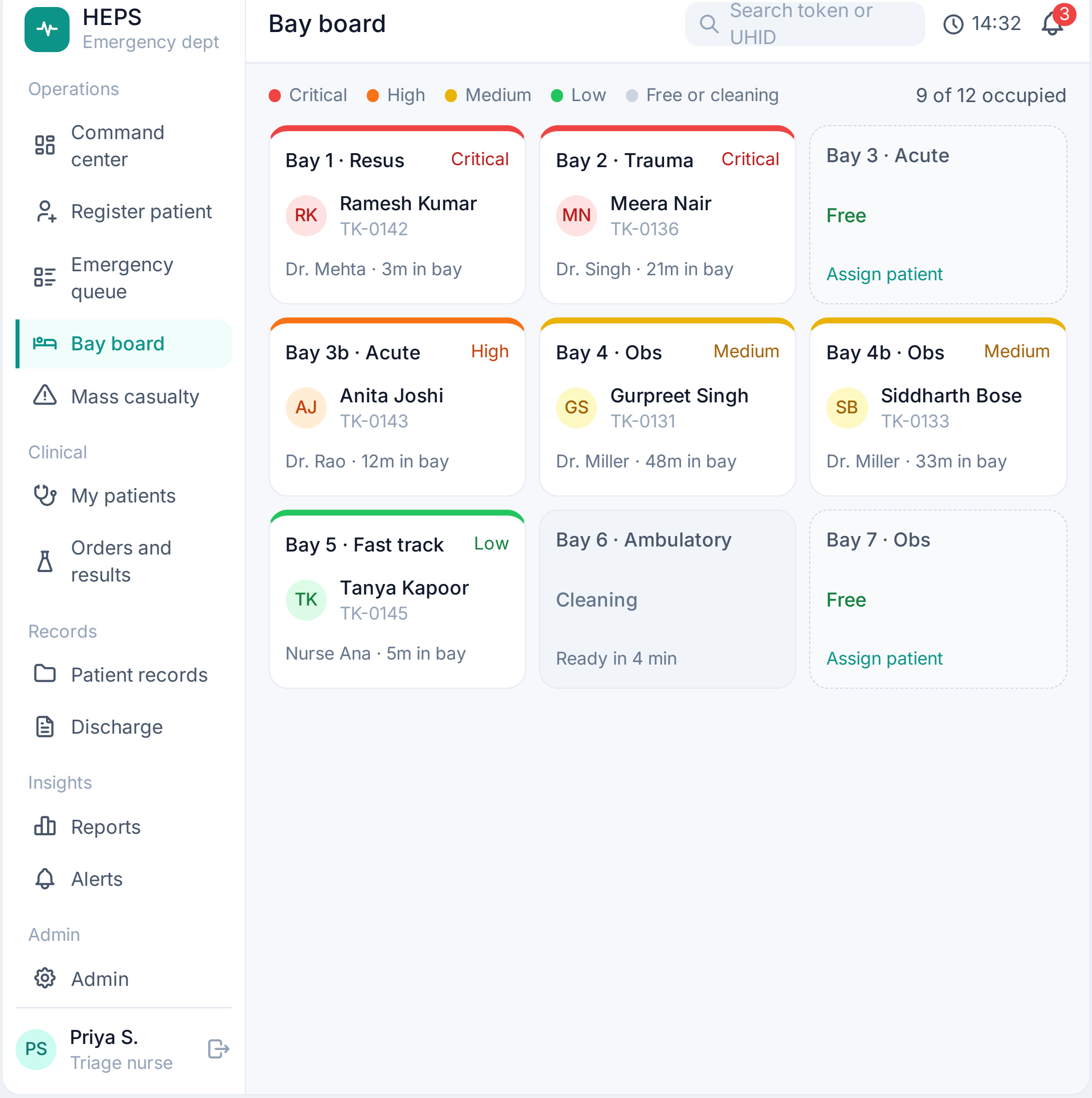
Build notes

- Table is ordered by PatientPriorityComparator: acuity, then escalated flag, then arrival time. Escalated patients show a small label with the reason on hover.
- Wait vs SLA bar: green under 75% of target, amber over 75%, red when breached.
- Selected row is highlighted in teal tint with a left accent. Bottom action bar: Re-triage, Print slip, Assign doctor, Begin treatment.
- Columns must never be clipped at 1280px; use ellipsis plus tooltip for long text. Add Mark as left without being seen as a row action.

08

Bay board

Assignment · visual ED bay and bed management



Build notes

- Each bay is a resource with type, status (Free, Occupied, Cleaning, Blocked) and current visit. Capacity is enforced: no double booking.
- Tile top border follows priority colour. Shows patient initials, token, doctor, minutes in bay. Free bays use a dashed border and an Assign patient link.
- Releasing a bay sets it to Cleaning for a configurable time, then Free. Critical bays get a soft pulsing red glow.
- Clicking a tile opens assign, transfer or release actions. Drag and drop is optional.

09

Doctor workspace

Treatment · patient page with journey stepper and tabs

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Triage nurse

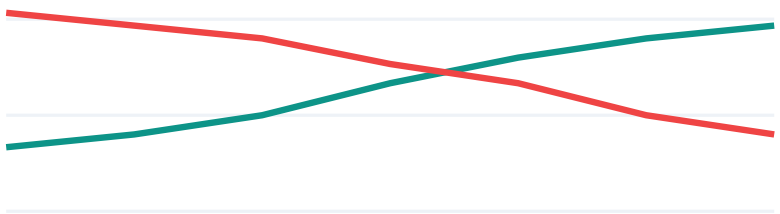
RK **Ramesh Kumar** 58 · Male
UH-2026-000891 · TK-2026-0142 · Bay 1 Resus
Critical
Allergy: penicillin
In treatment · 14m

Registered — Triaged — Assigned — Treatment — Disposition

Overview | Notes | Orders | Results | Vitals | Disposition

Vitals trend

last 30 min



SpO2: 88 to 94% HR: 134 to 108

Latest vitals

SpO2 94%	Heart rate 108
BP 140/90	NEWS2 6

Orders

Add order

12-lead ECG	Ordered 14:05	Resulted
Troponin I	Ordered 14:06 · result abnormal	In progress
Aspirin 300 mg, oral	Ordered 14:07	Ordered

Build notes

- Sticky patient banner on every patient page: name, age and sex, UHID, token, bay, priority pill, allergy in red, time in treatment.
- Journey stepper: Registered, Triaged, Assigned, Treatment, Disposition. Current step has a teal ring.
- Tabs: Overview, Notes (SOAP, signed and timestamped), Orders, Results, Vitals (add repeat set and re-run rule engine), Disposition.
- Vitals trend chart (SpO2 teal, HR red) with light grid. Orders list with status pills: Ordered, In progress, Resulted.

10

Orders and results

Treatment · lab, imaging, medication orders

HEPS
Emergency dept

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PS
Priya S.
Triage nurse

Orders and results

Search token or UHID

14:32

3

All | Pending | Resulted

Troponin I · Ramesh Kumar
Lab · ordered 14:06 · Bay 1
Abnormal

12-lead ECG · Ramesh Kumar
Cardiac · ordered 14:05 · Bay 1
Resulted

X-ray left leg · Anita Joshi
Imaging · ordered 14:12 · Bay 3
In progress

CBC · Kavita Sundaram
Lab · ordered 14:20 · Bay 4
Ordered

Result · Troponin I

resulted 14:31

Test	Value	Reference	Flag
Troponin I	0.42 ng/mL	under 0.04	High
CK-MB	8 ng/mL	0 to 5	High
Potassium	4.1 mmol/L	3.5 to 5.0	Normal

Add to notes | Acknowledge result

Build notes

- Order lifecycle: Ordered, In progress, Resulted. Abnormal results raise an alert and show a red Abnormal tag.
- Result panel shows value, reference range and a flag pill (High, Low, Normal). Doctor acknowledges the result, which is audited.
- Add to notes copies the result into the SOAP note. Orders are created from the patient Orders tab.
- Order types: lab tests, imaging, ECG, medications, IV fluids, procedures.

11

Disposition and discharge

Exit · outcome, diagnosis, medicines, PDF summary

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Emergency dept

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PS Priya S.
Triage nurse

Disposition · Gurpreet Singh

Search token or UHID

14:32

3

Outcome

- Discharged home
- Admit to ward
- ICU or OT
- Transfer out
- LAMA
- Referred

Discharge details

Diagnosis (ICD-10) *

I20.0 Unstable angina

Medicines on discharge

Aspirin 75 mg, once daily
Atorvastatin 20 mg, night

Follow-up date

29 Oct 2026

Advice

Rest, low-salt diet, return if pain recurs

Summary preview

PDF

City General Hospital
Discharge summary

Patient: Gurpreet Singh, 48 M
UHID: UH-2026-000855
Admitted: 22 Oct, 11:40 · Left: 14:55
Diagnosis: Unstable angina (I20.0)
Treatment: ECG, troponin, aspirin loading

Follow-up 29 Oct Dr. Mehta, signed

Print

Complete and release bay

Build notes

- Six outcomes as selectable tiles: Discharged home, Admit to ward, ICU or OT, Transfer out, LAMA, Referred. Required fields change per outcome (ward and bed for Admit, hospital name for Transfer).
- Diagnosis picker uses a seeded ICD-10 list. Medicines, follow-up date and advice are saved to the dispositions table.
- Summary preview is generated as a PDF (PDFBox or OpenPDF) with hospital header, diagnosis, treatment, medicines and doctor signature.
- Complete and release bay sets the visit to COMPLETED and the bay to Cleaning.

12

Patient records

History · search and visit history by UHID

Patient records

Search token or UHID · 14:32 · 3

Export CSV

UHID, token, name, phone · Priority · Status · Last 7 days

Patient	Priority	Status	Arrival
Ramesh Kumar UH-2026-000891 · 58 M	Critical	In treatment	22 Oct, 14:21
Anita Joshi UH-2026-000872 · 34 F	High	In treatment	22 Oct, 14:09
Gurpreet Singh UH-2026-000855 · 48 M	Medium	Discharged	22 Oct, 11:40
Sunita Reddy UH-2026-000849 · 39 F	High	Admitted	22 Oct, 10:55
Tanya Kapoor UH-2026-000901 · 12 F	Low	Waiting	22 Oct, 14:30

Ramesh Kumar · visit history

3 visits

- 22 Oct 2026 · Chest pain
  Critical · Cardiology · in treatment
- 02 Aug 2026 · Fever — Discharge summary
  Medium · General · discharged home
- 11 Jan 2026 · Minor cut — Discharge summary
  Low · Fast track · discharged home

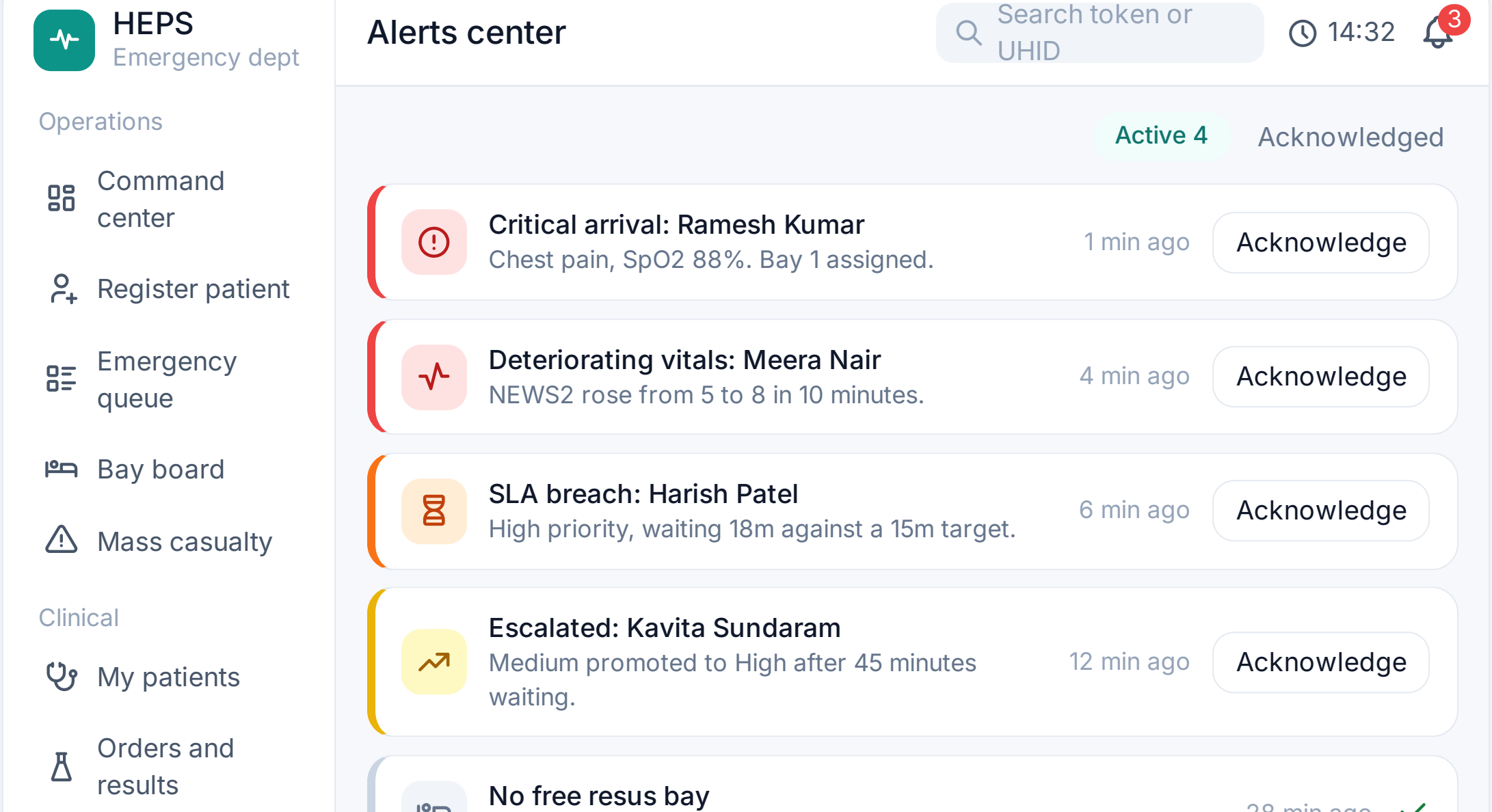
Build notes

- Search by UHID, token, name or phone. Filters: priority, status, department, doctor, date range. Phone numbers are masked unless the role allows.
- Status pills: In treatment (blue), Discharged (green), Admitted (violet), Waiting (amber).
- Selecting a patient shows the full visit history for that UHID with a link to each discharge summary.
- Export to CSV or PDF. Replaces the old Patient Records table.

13

Alerts center

Monitoring · real-time alert list



Build notes

- Alert types: critical arrival, SLA breach, escalation, deteriorating vitals, no free bay, abnormal result.
- Left border and icon tint follow severity (red, orange, yellow, grey). The bell in the top bar shows an unread count badge.
- Acknowledge records user and time in the audit log. Acknowledged rows fade and show a green check.
- Generated by the 15-second background engine and by service events (new vitals set, result posted).

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Reports and analytics

Insights · operational metrics

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PS Priya S.
Triage nurse

Reports and analytics

Search token or UHID

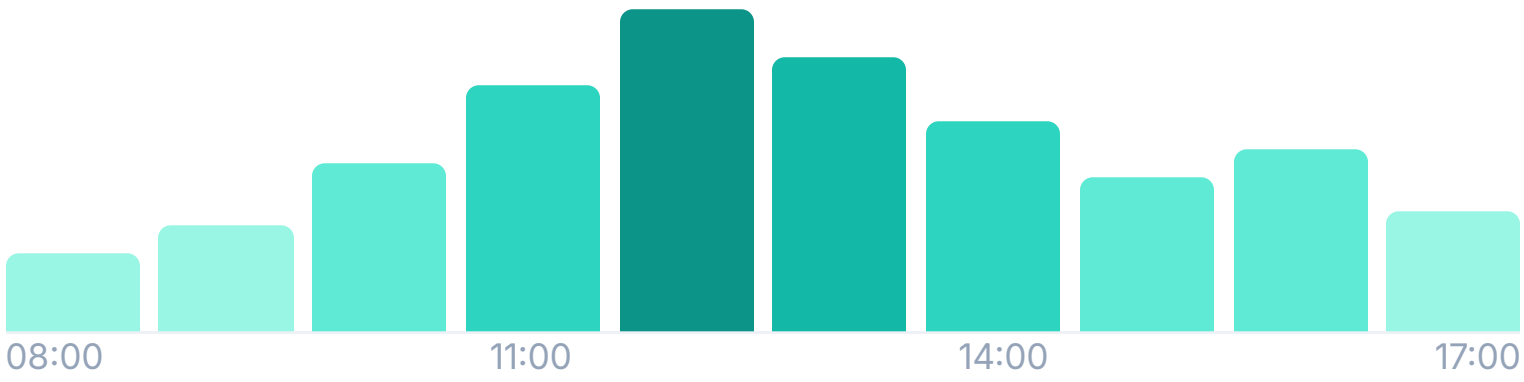
14:32

3

Today | 7 days | 30 days | Export PDF

Patients seen	Door to triage	Door to doctor	Left unseen
48	7m	24m	2%
up 8% vs yesterday	target 10m	target 20m	1 patient

Patients per hour



Acuity mix

Critical	12%
High	24%
Medium	40%
Low	24%

SLA compliance

91%

4 breaches today, 3 in the 16:00 hour

Build notes

- Metrics: patients seen, door to triage, door to doctor, left without being seen, acuity mix, SLA compliance, bay utilisation, doctor workload.
- Charts use light grid lines, rounded bars, teal gradient by volume, priority colours for acuity. White cards, no chart borders.
- Range switch: Today, 7 days, 30 days. Export to PDF or CSV. Visible to Admin and Charge nurse.
- Compute metrics in ReportService from visits timestamps (arrival, triage, doctor seen, disposition).

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Admin

Management · users, doctors, bays, settings

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PS Priya S.
Triage nurse

Admin

Search token or UHID

14:32

3

+ Add user

Users | Doctors | Bays | Hospital settings

Staff member	Role	Last login	Active
PS Priya S. nurse.priya	Triage nurse	Today, 13:58	
AM Dr. Arjun Mehta dr.mehta	Doctor	Today, 14:02	
RV Ravi V. desk.ravi	Receptionist	Today, 13:10	
AN Anita N. charge.anita	Charge nurse	Today, 08:00	
AD Administrator admin	Admin	Yesterday	

Passwords are stored hashed. Every change here is written to the audit log.

Build notes

- Tabs: Users, Doctors (specialty, shift, on or off duty), Bays (type, capacity), Hospital settings (name, logo, SLA thresholds, escalation minutes).
- Roles shown as coloured pills. Active toggle disables an account without deleting it.
- Audit log screen (Admin only) lists user, role, action, patient, old and new value, time.
- Hospital branding from settings is used on login, top bar, triage ticket and PDFs.

16

Mass casualty mode

Special mode · START triage and bulk registration

HEPS
Emergency dept

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PS Priya S.
Triage nurse

Mass casualty

Search token or UHID

14:32

3

Mass casualty mode is on
Simplified START triage. Started 14:40 by Charge nurse Anita.

End mode

Red · immediate	Yellow · delayed	Green · minor	Black · deceased
7	11	16	1

Quick registration

Name or "Unknown male" | Age | Sex

Red | Yellow | Green | Black — One tap tags and registers

Tag	Patient	Location	Time
Red	Unknown male, about 40	Bay 1 · Resus	14:43
Red	Priyanka D., 27	Bay 2 · Trauma	14:44
Yellow	Mohan L., 52	Corridor A	14:46
Green	Unknown female, about 20	Waiting hall	14:47

Build notes

- Toggle for Admin and Charge nurse. Shows a light red banner (not dark) with who started it and when.
- START colour tags: Red immediate, Yellow delayed, Green minor, Black deceased. One tap registers and tags with only name or Unknown, age and sex.
- Non-essential fields are hidden. Bulk tag list shows tag, patient, location and time. Full details can be completed later.
- Ending the mode returns to normal triage; tagged patients map to Critical, High and Low.

Workflow and role access

Patient status pipeline

Registered → Triaged → Waiting → Assigned (bay + doctor) → In treatment → Observation or results pending → Disposition → Completed

Enforced by a PatientStateMachine. Invalid transitions are rejected and logged. The pipeline strip on the Command center and the journey stepper on the patient page both read from this status.

Who sees which screen

Screen	Receptionist	Triage nurse	Doctor	Charge nurse	Admin
Command center	•	•	•	•	•
Register patient	•	•	–	–	•
Emergency queue	–	•	•	•	•
Bay board	–	•	–	•	•
My patients and workspace	–	–	•	–	•
Orders and results	–	•	•	•	•
Disposition and discharge	–	–	•	•	•
Patient records	•	•	•	•	•
Reports and analytics	–	–	–	•	•
Alerts center	–	•	•	•	•
Admin and audit log	–	–	–	–	•
Mass casualty mode	–	–	–	•	•

Audit rule

Register, triage override, assign, begin treatment, order, result acknowledge, disposition, print, login and logout, user and settings changes all write a row to audit_log with user, role, action, patient, old value, new value and time.